Batch And Scanner Effects In Election Forensic Analytics: A Method Framework For Investigative, Non-Certifying Group-Level Signals

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Abstract

Forensic analytics prioritize investigation. They do not prove fraud, certify outcomes, or replace risk-limiting audits. This paper defines the W.03 method framework for batch and scanner effects: systematic deviations associated with a tabulation batch, scanner, drop box, or upload. The signal is group-level, not a claim about an individual ballot. The method models vote share or residuals with a batch/scanner grouping factor, estimates group-level fixed or random effects, and flags groups whose effects are large relative to within-group variation. RCOUNT batch manifests, cast vote records, and summaries provide the intended hash-bound inputs. Reports instantiate the W.01 forensic report contract with `model_id=batch-effect`, source package hashes, unit-level scores, and plain caveats. This is a method and framework paper: the report contract is specified in W.01, the statistics are established, and no forensics crate, fixture, function, or test is claimed to exist.

1 Introduction

Forensic analytics prioritize investigation. They do not prove fraud, certify outcomes, or replace risk-limiting audits. A batch/scanner effect is a systematic deviation associated with a tabulation batch, scanner, drop box, upload, or similar operational grouping. It is a grouping-level signal: the question is whether units processed through the same group move together after ordinary comparisons are accounted for, not whether any single ballot is suspicious.

This boundary also separates the W-series from the V-series. V-series work replays audit or canvass evidence and can support certification-style claims when its assumptions are met. W-series work produces investigative analytics: outlier scores, residuals, digit checks, batch effects, spatial anomalies, and discrepancy clustering. The output is an investigation queue with caveats, never a pass/fail certification result.

Batch and scanner effects are a natural W-series family because election data often arrive in operational groups. Vote-by-mail batches may differ from election-day batches. Early-vote uploads may differ from precinct uploads. Scanners may serve different locations, times, or ballot styles. Some group differences are therefore real and innocent. The method in this paper is meant to identify groups whose deviations deserve review while preserving that interpretive boundary.

2 Method

The method models reported results with a batch/scanner grouping factor. The response can be a vote share, a transformed vote share, or a residual from a baseline model that already accounts for contest, jurisdiction, ballot style, mode, time, or comparable historical information. The grouping factor is the batch, scanner, drop box, upload, or other operational unit being evaluated.

A fixed-effect version estimates a separate coefficient for each group. A random-effect version treats group effects as draws from a common distribution and estimates the degree to which each group departs from the population of comparable groups. Either form asks the same forensic question: is the estimated group effect large relative to the variation observed within and across groups? A report may rank groups by standardized effect size, posterior or empirical shrinkage estimate, residual dispersion, or a calibrated tail probability.

The score is not a fraud finding. Legitimate operational differences routinely produce real batch effects. Vote-by-mail batches can have different candidate shares than in-person precinct batches. Early batches can differ from election-day batches. A scanner assigned to one geography or ballot style can inherit that geography or style. For that reason, the preferred baseline makes the comparison population explicit and includes the operational variables needed to avoid treating ordinary election administration as an anomaly.

The method flags groups whose effect remains large after that comparison. It also records the scale on which the score was computed, the baseline population, and an investigation note describing the most likely benign grouping explanation when one is visible from the inputs. The statistical machinery is established; this paper fixes the W-series framing and report shape for this analytic family.

3 RCOUNT Inputs And Report Contract

The intended inputs are RCOUNT batch manifests, cast vote records, and normalized summaries. Batch manifests identify the operational grouping: batch, scanner, drop box, upload, or equivalent source. Cast vote records and summaries provide the contest totals or ballot-level records from which vote shares and residuals are derived. Source package hashes bind the report to the exact RCOUNT package used to compute it.

W.03 instantiates the W-series report contract defined in W.01. The per-record fields are:

Field	W.03 value or meaning
method_id	Registered batch/scanner-effect analytic.
feature_set	Variables used for vote share, residuals, grouping, and baseline adjustment.
baseline_population	Comparable contests, units, modes, times, or groups used for calibration.
model_id	batch-effect.
unit_id	Batch, scanner, drop box, upload, or other operational group.
score	Group-level anomaly score or standardized effect estimate.
p_value_or_rank	Optional calibration or rank within the baseline population.
investigation_note	Caveat explaining the grouping, comparison, and plausible benign boundary.

The report is about the group, not individual ballots. A scanner effect should be attributed to the scanner only when the inputs support that grouping. A mail batch effect should not be described as a precinct effect merely because the batch contains precinct-coded records. The central discipline is attribution: identify the level at which the deviation is measured, bind it to source package hashes, and keep the result in the W-series investigative transcript rather than V-series certification evidence.

4 Intended Fixture Pattern

The fixture family is intended, not claimed as implemented. It follows the four-role pattern from W.01 and adapts it to batch and scanner effects.

Fixture	Role
No-anomaly	A synthetic package whose groups differ only by ordinary noise after the declared baseline. A correct method returns low scores and no urgent queue.
Single planted anomalous batch	One batch or scanner is given a planted group-level deviation. The method should rank that group near the top and attribute the signal to the planted grouping.
Real-but-innocent batch effect	A package contains a legitimate operational split, such as mail versus in-person voting. The method may detect the effect, but the report note must present it as an expected process difference rather than proof.
False-positive boundary	A package is unusual under a weak baseline but explainable under the correct grouping or comparison population. The method should expose the sensitivity and keep the caveat attached to the score.

These roles test two requirements at once. The method should find a planted group-level deviation when the data support one. It should also avoid promoting a legitimate operational difference into an allegation. The fixtures therefore exercise scoring, attribution, baseline disclosure, and the investigation-note field, not just numerical detection.

5 Boundaries

Batch effects are expected in real election administration. Different modes, times, geographies, ballot styles, and processing streams can all create group-level differences. A correlation with a scanner or batch is therefore not proof that the scanner or batch caused the pattern, and it is not proof of fraud. It is a reason to check records, procedures, chain of custody, and ordinary operational explanations.

The method is descriptive rather than causal. It identifies a grouping-level association after a stated comparison, but causation belongs to a separate investigation. A weak baseline can create false positives by comparing mail batches to election-day precincts, early uploads to final uploads, or scanners serving different ballot styles. The baseline population and feature set are therefore part of the result, not background metadata.

The lead boundary governs every report: forensic analytics prioritize investigation; they do not prove fraud, certify outcomes, or replace risk-limiting audits. W.03 reports must remain separate from V-series evidence and must carry their caveats with their scores.

6 Conclusion

Batch and scanner effects are useful forensic signals because many election records are organized by operational grouping. They are also easy to overread. This paper defines the W.03 method framework for treating those effects as investigative, group-level deviations: model vote share or residuals with a batch/scanner factor, estimate group-level effects, rank groups relative to within-group and baseline variation, and report the result through the W.01 contract.

The contribution is not a shipped implementation. No forensics crate, fixture, function, or test is asserted here. The contribution is the method boundary and report discipline: bind the analysis to RCOUNT source hashes, attribute the score to the correct grouping, disclose the baseline, and state plainly that the output is an investigation queue rather than proof or certification.

References

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